Secure Purview Starter Kit

Implementation Principle

Build a foundation that people can understand and operate. Add protection in deliberate layers, at a speed the organization can absorb, while moving forward without leaving administrators or users behind.

Disclaimer

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Use this material for implementation planning. Microsoft products, licenses, services, and management interfaces can change. Check the design against current Microsoft documentation. Also check it against your environment, laws, privacy needs, data-handling rules, and risk tolerance.

Test every script in a non-production tenant first. Before production use, obtain approval from the right security, compliance, privacy, legal, operations, and business owners. Export the current configuration. Prepare and document a rollback plan.

Summary: A Safe Starting Point

The Purview Starter Kit is a safe springboard for organizations that are starting their data-protection journey. It creates a practical foundation for information protection and data loss prevention, or DLP. The goal is not to enable every Microsoft Purview solution at once. The goal is to start with a set of labels and a manageable set of DLP controls. Users can learn how to classify information. Administrators can learn how to monitor, investigate, support, and tune the controls. The organization can then add more protection based on real evidence.

Start with Import-PurviewBaseline.ps1. This script creates the basic label structure and publishes the labels. It configures mandatory labeling, default labels, the first DLP policies, and the global Endpoint DLP settings. It also configures supporting features when the required licenses and interfaces are available. Next, add the lists of personal webmail and cloud-storage destinations. Run Deploy-CT-EDLP-EgressControl.ps1 only after those lists exist. CT-EDLP is the name this kit uses for its set of endpoint DLP egress controls. The egress policies depend on the labels and destination groups created earlier.

This order reduces risk. It prevents incomplete or overly broad Endpoint DLP rules. It also limits the amount of change introduced at one time. Users can first learn what the labels mean.

Administrators can first learn how to review events, investigate matches, answer questions, and tune policies. This makes the later controls easier to operate.

What the Starter Kit Is, and What It Is Not

The Starter Kit Is	The Starter Kit Is Not
A tested starting point for labels, label publishing, DLP, and gradual expansion.	A final design for every organization. It does not replace legal, privacy, risk, or business decisions.
A way to reduce the effort needed to start. It still leaves clear places for tuning.	A plan to deploy the full Microsoft Purview stack in one project.
A step-by-step path that builds knowledge before enforcement.	A reason to enable broad blocking before the impact and dependencies are understood.
A repeatable baseline that each organization can adapt.	A one-size-fits-all design that must be copied without change.

Foundation: Import-PurviewBaseline.ps1

Import-PurviewBaseline.ps1 creates the starting point. The labels give users and administrators a shared language for data sensitivity. The script publishes the labels to users. It also creates the first DLP policies and configures the global Endpoint DLP settings. When the required licenses and interfaces are available, it can also configure message encryption, Insider Risk Management objects, and Adaptive Protection.

Every organization needs to adapt the taxonomy to its own needs. One company may not need all the labels in the baseline. Another may need additional labels for more granular controls. Always aim for the smallest number of labels that still meets your requirements. Fewer labels are more intuitive, and users can select the right one more easily. Some organizations are well served by just three labels: Public, General, and Confidential. You can always add labels later, but removing them once users have learned them is much more difficult. The labels in the baseline are suggestions based on experience in many customer environments, not a requirement.

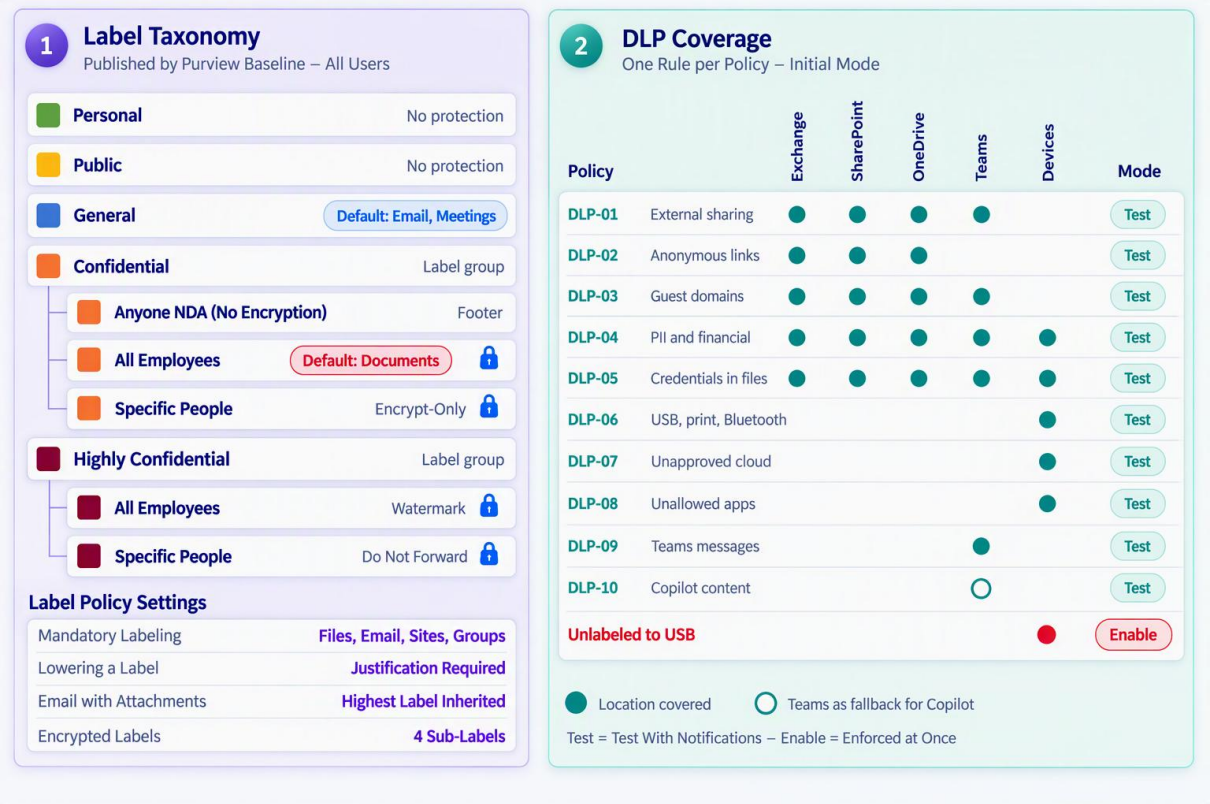


Figure 1: Foundation outcome after Import-PurviewBaseline.ps1. The label taxonomy and initial DLP portfolio provide a shared starting point for users and administrators.

Build Skills Before Adding More Controls

For a new implementation, use the baseline before adding the detailed egress controls. Give users time to learn the labels. They need to understand the difference between Personal, Public, General, Confidential, and Highly Confidential information. They also need to see how encryption, markings, and sharing restrictions affect daily work. Administrators need hands-on experience too. They must learn where labels appear in Office, SharePoint, OneDrive, and Teams. They must know how to use Activity Explorer, investigate DLP matches, handle exceptions, and tune policies.

> **Do Not Confuse Progress with Volume**
>
> Deploying more Purview solutions does not automatically create a more mature information-protection program. A smaller set of well-understood, owned and measured controls is safer than a large set of controls that users cannot predict and administrators cannot operate.

Taxonomy Is a Starting Point, not a Fixed Corporate Truth

The starter taxonomy is deliberately small. This makes it easier to learn and support. Use it as provided when it fits your organization. Change it when your business language, data types, collaboration model, or regulatory duties are different. You can adjust names, descriptions,

sublabels, defaults, encryption, markings, container settings, and publishing scopes. There is no single taxonomy that fits every company. Keep the final design simple. Labels should be clear and should not overlap unless there is a good reason.

Design Question	Why It Matters
Do the label names match the terminology people already use?	Familiar words help users choose the correct label.
Which labels require encryption, and for whom?	Encryption can affect guests, external recipients, applications, and recovery procedures.
Should documents receive an encrypted default label?	A protected default can reduce risk. It can also disrupt external sharing and automated workflows.
Which populations genuinely need different publishing settings?	Create another publishing policy only when it solves a clear business need.
Which legal or regulatory categories require dedicated controls?	Legal or regulatory needs may require more labels, sensitive information types, exact data match, or classifiers.

Extension Layer: Destination Intelligence and CT-EDLP for Endpoint Protection

Labels classify information, but classification alone does not stop data from leaving the organization. Corporate Windows 11 endpoints are where people open, copy, print, upload, and send sensitive files every day. They are therefore a main point where data can leave the managed environment, whether by mistake or on purpose. Endpoint protection closes this gap. It uses the sensitivity label on a file to decide what a user may do with it on the device. The visual below shows the data-exit paths that the starter kit controls: device actions, browser and cloud uploads, applications, and external email. The purpose is to protect sensitive data from being egressed from corporate Windows 11 endpoints, while allowing normal business work to continue.

The key prerequisite is that the endpoints must be onboarded to Microsoft Purview. Endpoint DLP policies are enforced by the device itself, not in the cloud. Onboarding makes the device known to Purview and enables its endpoint DLP capabilities. Only then can the device detect file activities such as copy, print, upload, and paste. It can also check the label of the file involved, apply the policy, show policy tips to the user, and report events to Activity Explorer and alerts. A device that is not onboarded does not enforce the policies and does not produce any evidence. The CT-EDLP controls would then appear to be deployed while the endpoint stays unprotected. Confirm onboarding for the pilot devices before you deploy the egress controls and expand it to all in-scope devices before you move to enforcement.

How to Onboard Windows Devices to Microsoft Purview

Device onboarding is shared between Microsoft Purview and Microsoft Defender for Endpoint. Devices that are already onboarded to Defender for Endpoint appear in the Purview device list, and no further steps are needed for them. Onboarding a device in the Purview portal also onboards it to Defender for Endpoint. If another antivirus product is the primary one, Defender for Endpoint runs in passive mode, so it does not interfere with the existing protection.

Onboard Windows devices into Microsoft 365 overview

Before you start, check these points:

- The devices run a supported version of Windows 11, with a current anti-malware client version.
- Your account is a member of the Security admin, Compliance admin, or Global admin role. Use the role with the fewest permissions that works.
- The required licenses are in place.

Then onboard the devices:

1. Open the Microsoft Purview portal and go to Settings, then Device onboarding, then Devices. Select Turn on device onboarding.
2. Go to Onboarding. Select the operating system and the deployment method that fits your environment, for example Microsoft Intune, Group Policy, Microsoft Configuration Manager, or a local script. Download the onboarding package.
3. Deploy the package with your device-management tool, first to a pilot group of test devices.
4. Verify the result. The devices must appear in the Purview device list and report activity. Fix any devices that do not appear before you continue.
5. Expand onboarding to all in-scope devices before you move any CT-EDLP control from simulation to enforcement.

What the Extension Scripts Add

Three extension scripts in the Secure Purview Starter Kit add the endpoint protection layer. Add-WebmailDomainsToDlpGroups.ps1 creates and fills two groups for personal webmail services. Add-CloudStorageDomainsToDlpGroups.ps1 creates and fills two groups for cloud-storage services that the organization does not approve. Deploy-CT-EDLP-EgressControl.ps1 then creates or updates eleven egress policies. These policies cover device actions, browser and cloud uploads, applications, and external email.

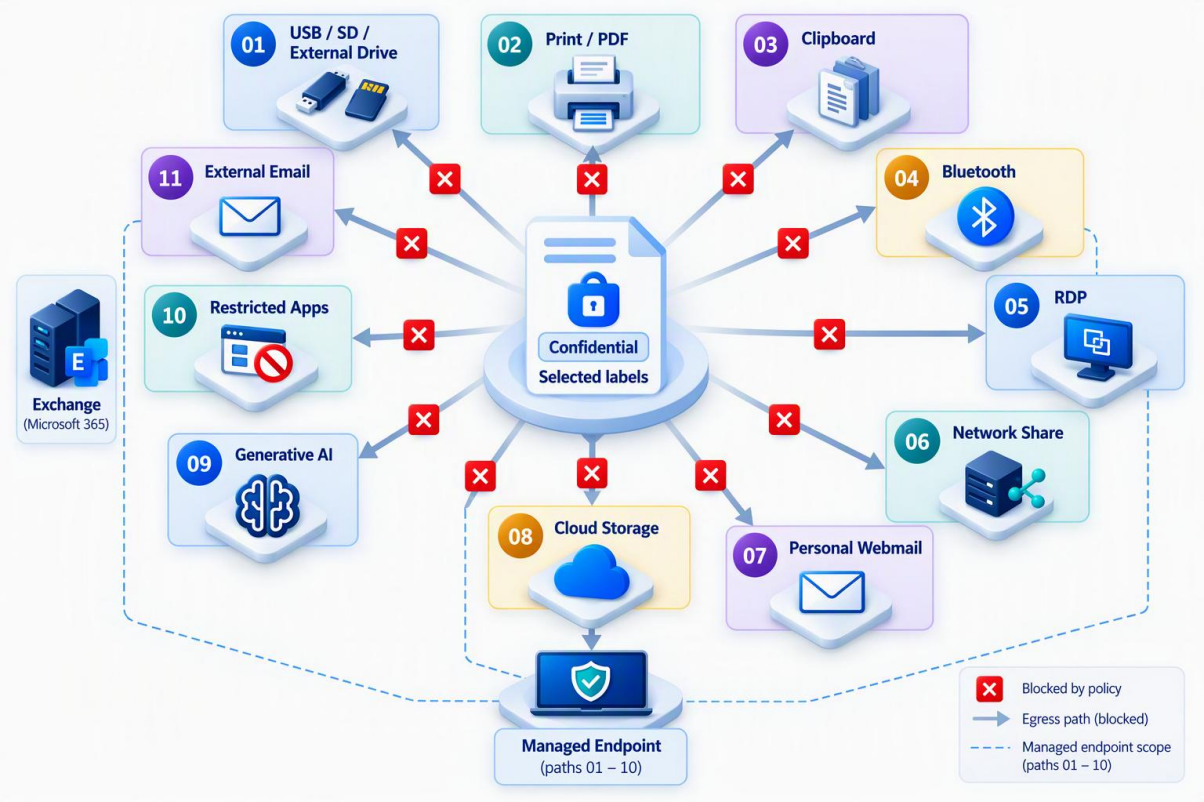


Figure 2: Egress paths protected on a corporate Windows 11 endpoint. Labeled, sensitive files on an onboarded device are controlled across four groups of exit paths: six device actions (removable storage, printing, clipboard, Bluetooth, remote desktop redirection, and network-share copying), three browser and cloud uploads (personal webmail, unapproved cloud storage, and generative AI destinations), restricted applications, and external email. The eleven CT-EDLP controls evaluate the sensitivity label together with the action or destination. The device must be onboarded to Microsoft Purview so that the controls are enforced and their events are reported.

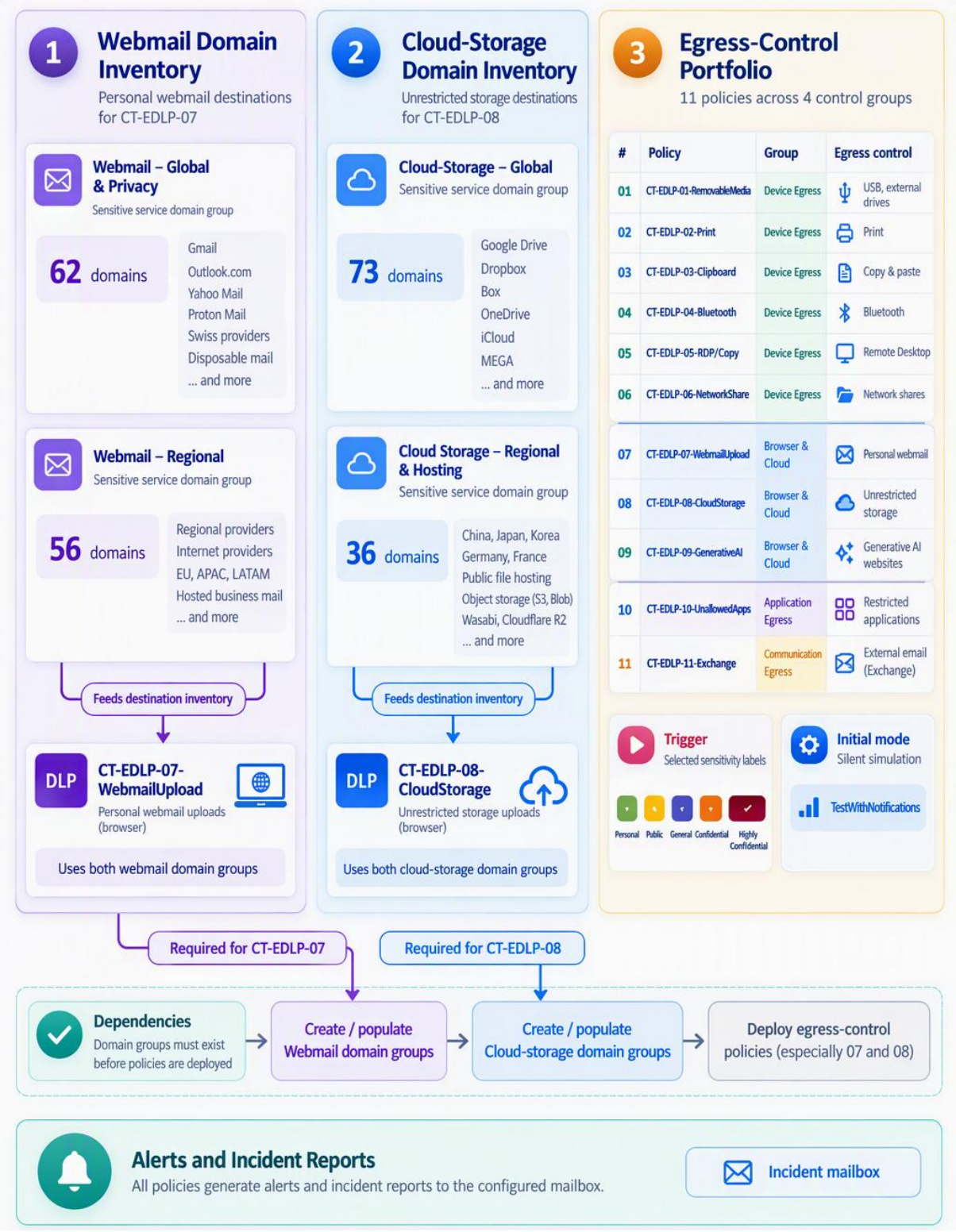


Figure 3: Outcome of the three extension scripts. The webmail and cloud-storage inventories are dependencies for CT-EDLP-07 and CT-EDLP-08; the deployment utility adds the eleven channel-specific controls.

Why Sensitive Service Domain Groups Are Needed

An egress control must know where data is going. A rule that blocks uploads to personal webmail or unapproved cloud storage needs a reliable list of those destinations. In Microsoft Purview, sensitive service domain groups provide this list. Each group is a named collection of web domains that Endpoint DLP rules can reference. The rules then act on the group instead of on individual website addresses.

The starter kit uses four groups. Two cover personal webmail services, one global and one regional. Two cover cloud-storage services, one global and one regional. The domains come from text files that list the services in each category. The lists are maintained and will be updated regularly, because new services appear and existing ones change their domains. Re-running the domain scripts updates the groups from the latest lists. The policies keep pointing to the same groups, so you do not need to edit the policies or rules when a list changes.

Keeping the groups separate also keeps the controls precise. Webmail and cloud storage are different risks and may need different rules, exceptions, and user guidance. Separate groups let CT-EDLP-07 act only on personal webmail and CT-EDLP-08 act only on unapproved cloud storage. Without these groups, the rules would have no defined destinations, or they would have to contain long, hard-to-maintain domain lists. That is why the group scripts run before Deploy-CT-EDLP-EgressControl.ps1. Review each list before you import it and remove any service your organization approves for business use.

What the Endpoint Protection Scripts Deploy

Script	Resulting Objects	Purpose
Add-WebmailDomainsToDlpGroups.ps1	Webmail - Global & Privacy: 62 domains; Webmail - Regional: 56 domains	Creates the personal-webmail destination list used by CT-EDLP-07-WebmailUpload.
Add-CloudStorageDomainsToDlpGroups.ps1	Cloud Storage - Global: 73 domains; Cloud Storage - Regional & Hosting: 36 domains	Creates the unapproved cloud-storage destination list used by CT-EDLP-08-CloudStorage.
Deploy-CT-EDLP-EgressControl.ps1	11 policies and 11 matching rules across four control groups	Creates label-based controls for specific data-exit paths. Start them in silent simulation unless another mode has been approved.

The Combined Outcome After All Four Scripts

After all four scripts run, the tenant has two connected protection layers. The baseline layer defines how information is classified. It publishes the labels and provides the first DLP coverage. The extension layer adds known destination lists and label-based egress controls. Together, the layers answer four questions: What type of information is this? How did the user classify it? Where is the information going? What action is the user trying to perform?

Required Order and Dependencies

The order below describes a true greenfield approach, where no label taxonomy exists yet. Your organization may already have a sensitivity label taxonomy and wants to add only endpoint protection with DLP. In that case, you can go directly to step 4. Then work with your existing label taxonomy in step 6, when you select the trigger labels for the CT-EDLP policies. Steps 1 to 3 are not needed but confirm that your existing labels are published and that the support and operations teams understand them.

Order	Action	Dependency and Reason
1	Review and adapt the target taxonomy	Confirm the label language, protection behavior, primary domain, licenses, endpoint readiness, and support model before making tenant changes.
2	Run Import-PurviewBaseline.ps1	The labels must exist before the later policies can use them. This step also gives users and administrators a common foundation for learning.
3	Operate the greenfield baseline	Use the baseline for a period of time. Answer user questions. Review DLP events. Tune the policies. Confirm that the support and operations teams can explain and investigate what happens.
4	Run Add-WebmailDomainsToDlpGroups.ps1	Creates the two destination groups used to identify personal-webmail uploads.
5	Run Add-CloudStorageDomainsToDlpGroups.ps1	Creates the two destination groups used to identify unapproved cloud-storage uploads.

Order	Action	Dependency and Reason
6	Run Deploy-CT-EDLP-EgressControl.ps1	The script checks the labels and destination groups. It then creates or updates the selected policies and rules. Keep the policies in silent simulation until enforcement is approved.
7	Validate, tune and stage enforcement	Review Activity Explorer, alerts, incident reports, destination coverage, policy overlap, application dependencies, and business impact. Enable controls only for approved pilot groups and channels.

Why This Order Matters

Labels come first. They are the language users see, and they are the trigger for many later controls. If labels keep changing, users cannot learn them and administrators cannot test the controls reliably.

The webmail and cloud-storage controls depend on different destination groups. Create those groups before deploying the related policies. Missing groups can make rules too broad or remove the intended separation between destinations.

Simulation results are useful only when the basic dependencies are ready. Endpoints must be onboarded. Labels must be published. Destination groups must contain the right domains. The operations team must know where to review matches and alerts.

Do not introduce classification, encryption, baseline DLP, Insider Risk Management, destination controls, restricted applications, artificial intelligence (AI) controls, and broad enforcement at the same time. Too many changes make problems hard to diagnose. Users also face too many new behaviors at once.

A phased approach creates clear checkpoints. Explain each layer. Test it. Tune it. Assign an owner. Add the next layer only when the current layer works and the organization can support it.

Adoption Principle: Build Capability as Protection Grows

Move at the fastest speed the organization can safely support. Script execution is not the measure of success. Users need simple label guidance, examples, and support. The service desk needs troubleshooting steps. Compliance and security teams need investigation procedures. Application and collaboration owners need a clear way to report business impact. Decision makers need simulation evidence before they approve enforcement.

The goal is steady progress and deliberate learning. Pause when the taxonomy, training, destination lists, or alert routing need correction. A short pause is not failure. It can prevent a larger rollback and protect trust in the data-protection program.

Recommended Rollout Model

Phase	Focus	Exit Criteria
Observe	Publish the baseline labels. Run the first DLP policies in test modes. Train users and administrators.	Users understand the labels. Support material is ready. Activity Explorer contains data. Major false positives and workflow gaps are known.
Prepare	Populate the webmail and storage groups. Review important entries. Confirm endpoint, AI, restricted-app, and incident-mailbox readiness.	The domain groups exist and have been reviewed. Dependencies and exceptions are documented. Current exports and rollback information are available.
Simulate	Deploy the CT-EDLP controls in TestWithoutNotifications. Run controlled test cases.	Targeted and non-targeted content behave differently as expected. Alerts and reports are useful. Policy overlap is understood.
Guide	Use TestWithNotifications for selected controls and pilot users.	Users understand the policy tips. Support processes work. Business owners accept the impact.
Enforce	Enable approved controls in stages. Start with selected channels and user groups.	Each control meets technical, operational, and business acceptance criteria. Monitoring and rollback remain ready.
Extend	Add auto-labeling, exact data match, classifiers, more DLP, Insider Risk Management tuning, or other Purview features only when there is a clear need.	Each new capability has a purpose, an owner, a test method, and measurable success criteria.

Operational Safeguards

Before each phase, export the current labels, label policies, DLP policies, rules, and Endpoint DLP settings.

- Use approved test accounts, synthetic data, onboarded test devices, and controlled destinations. Do not use real credentials or production-sensitive data for testing.
- Review the webmail and cloud-storage lists before import. Remove approved business services. Carefully review broad object-storage services.
- Run both domain scripts in preview mode first. Keep their backups and verification output. Confirm that each group is linked to the correct rule.
- Start CT-EDLP policies in silent simulation. A policy is not ready for enforcement just because the object was created successfully.
- Test CT-EDLP-11 carefully. Its effective Exchange condition can be broader than the intended label-based design.
- Confirm Restricted App Groups, generative AI readiness, and licensing before relying on CT-EDLP-09 or CT-EDLP-10.
- Assign clear owners for alerts, incident reports, user guidance, exceptions, policy tuning, and rollback decisions.

Implementation Decision Checklist

Decision	Owner	Required Before
Approve the sensitivity taxonomy and label descriptions	Information governance / compliance	Baseline deployment
Approve default labels, encryption and external collaboration behavior	Security, collaboration and legal	Baseline deployment
Confirm licensing, permissions, endpoint onboarding and support ownership	Technical service owners	Baseline deployment
Approve webmail and cloud-storage domain inventories	Security and application owners	Domain import
Approve trigger labels and CT-EDLP policy selection	Information governance and security	CT-EDLP deployment
Approve policy-tip wording, incident mailbox and response process	Security operations and service desk	Notification simulation
Accept simulation results and enforcement sequence	Business, security, compliance, privacy and legal	Enforcement

Conclusion

The Purview Starter Kit creates a durable starting point. It is not an instant end state. Import-PurviewBaseline.ps1 establishes the shared label and DLP foundation. The domain-group scripts add controlled destination lists. Deploy-CT-EDLP-EgressControl.ps1 then adds protection for specific channels and actions.

This order reduces technical risk and organizational overload. Users gain confidence with labels. Administrators build the skills needed to monitor, investigate, support, and tune DLP. The organization gathers evidence and adapts the controls to its own needs. The starter kit is therefore a safe springboard. It helps a company begin data protection with a solid foundation and grow at a sustainable pace.

References

The following references provide deeper product guidance and beginner learning. Use the current Microsoft documentation when you validate the design and prepare a production deployment.

AzFrameworks/PurviewBeginnersJourney

A beginner-friendly introduction to the Microsoft Purview ecosystem. It explains data governance, data security, and data compliance in a connected learning journey. It is useful before and during the starter-kit implementation.

Microsoft Learn – Purview Sensitivity Labels & Encryption

- Introduction to secure by default with Microsoft Purview
- Learn about sensitivity labels
- Get started with sensitivity labels
- Create and publish sensitivity labels
- Restrict access to content by using sensitivity labels to apply encryption
- Automatically apply a sensitivity label to Microsoft 365 data
- Use sensitivity labels to protect collaborative workspaces (groups and sites)
- Enable sensitivity labels for files in SharePoint and OneDrive
- Configure a label to apply S/MIME protection in Outlook
- Configure super users for Azure Rights Management | Microsoft Learn
- Learn about the Azure Rights Management encryption service
- What is Double Key Encryption (DKE)?
- Set up Double Key Encryption
- Sensitivity Labels in Power BI - Microsoft Purview Information Protection
- Sensitivity labels for Microsoft Teams

Microsoft Learn – Data Loss Prevention

- Learn about data loss prevention
- Plan for data loss prevention (DLP)
- Design a data loss prevention policy
- Data Loss Prevention policy reference
- Create and deploy a data loss prevention policy
- Data loss prevention Exchange conditions and actions reference
- Data Loss Prevention policy tips reference
- Use sensitivity labels as conditions in DLP policies
- Data loss prevention and Microsoft Teams
- Help prevent sharing sensitive items via SharePoint and OneDrive with external users

Microsoft Learn – Endpoint DLP

- Learn about Endpoint data loss prevention

- Get started with endpoint data loss prevention
- Configure endpoint data loss prevention settings
- Endpoint DLP policy scenarios in Microsoft Purview
- Help prevent exposure of sensitive files by configuring auto-quarantine for OneDrive sync
- Help prevent sharing of sensitive items with unauthorized cloud apps and services
- Onboard Windows 10 and Windows 11 devices into Microsoft Purview overview
- Onboard macOS devices into Microsoft Purview
- Microsoft Defender for Endpoint Device Control Removable Storage Access Control
- Learn about Advanced Label Based Protection

Microsoft Learn – Insider Risk Management

- Learn about Insider Risk Management
- Learn about Insider Risk Management policy templates
- Get started with Insider Risk Management
- Configure policy indicators in insider risk management
- Learn about Adaptive Protection in data loss prevention
- Help dynamically mitigate risks with Adaptive Protection
- Share Insider Risk Management data with other solutions

Microsoft Learn – Sensitive Information Types & Classification

- Learn about sensitive information types
- Sensitive information type entity definitions
- Learn about credential sensitive information types
- Learn about trainable classifiers
- Learn about exact data match based sensitive information types